f is a func. def. on $\mathbb{R}$

$\forall \varepsilon > 0, \exists \delta > 0$, s.t. $\forall x, y \in \mathbb{R}$,

$|x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$

↓(neg.)

$\exists \varepsilon > 0$, s.t. $\forall \delta > 0$, $\exists x, y$ s.t. $|x-y| < \delta$ and $|f(x)-f(y)| \geq \varepsilon$

Divisibility

Def.: Let $m, n \in \mathbb{Z}$, $n \in \mathbb{R}$ and $n \neq 0$,

Then n divides m → (notation: $n \mid m$)

$\Leftrightarrow \exists k \in \mathbb{Z}$, s.t $m = kn$

$\forall n \neq 0$, $n \mid 0$ is true

(Thm. (5))

$\forall a, b, m, n, d \in \mathbb{Z}$ and $d \neq 0$,

$\begin{cases} d \mid a \\ d \mid b \end{cases} \Rightarrow d \mid ma + nb$

Proof: Let $a, b, m, n, d \in \mathbb{Z}$ and $d \neq 0$

By def., $\exists k_1 \in \mathbb{Z}$, s.t. $a = k_1 d$

$\exists k_2$ — $b = k_2 d$

Then $ma + nb = (mk_1 + nk_2) d$ ← let j be

$m, k_1, n, k_2 \in \mathbb{Z} \Rightarrow j = mk_1 + nk_2 \in \mathbb{Z}$

By def., —

Thm.: $\forall m, n \in \mathbb{N}^*$ (>0), ($\to$ quotient) ($\to$ remainder)

$\exists$ unique $q, r \in \mathbb{Z}$,

s.t. $\begin{cases} m = qn + r \\ 0 \le r < n \ (\Leftrightarrow 0 \le r \le n-1) \end{cases}$

(i.e. $n \mid m \Leftrightarrow r = 0$)

Congruent

Def.: For $n \in \mathbb{Z}$ and $n > 0$ and $n \neq 1$

(Fixed)

① a is cong. to b modulo n

$\Leftarrow r_n(a) = r_n(b)$

(notation: $a \equiv b \pmod{n}$)

E.g.: ① $2 \equiv 12 \pmod{10} \equiv 22 \pmod{10} \equiv \cdots$

② $2 \equiv -8 \pmod{10}$

$(-8 = (-1) \cdot 10 + 2)$

Thm.: $a \equiv b \pmod{n} \Leftrightarrow n \mid b - a$

(By thm., ② 比較) ($\to$ ② $n \mid a - b$)

(=) Proof: (易證)

(反之亦然)

Let $n \mid a$, then $a \neq 0 \Rightarrow |a| \geq n$

By def., $\exists k \in \mathbb{Z}$, s.t. $a = kn$

$\Rightarrow$ Then $a \neq 0 \Rightarrow k \neq 0 \Rightarrow |k| \geq 1$

Then $|a| = |k| \cdot n \geq n$

(反之亦然)

Thus, $r_2 - r_1 = 0$

By def., $a \equiv b \pmod{n}$

Thm. : $\forall a,b,c,d \in \mathbb{Z}$ and $n \in \mathbb{R}$ and $n>0$, (∀)

① $\begin{cases} a \equiv b \pmod n \\ c \equiv d \pmod n \end{cases} \Rightarrow \begin{cases} a+c \equiv b+d \pmod n \\ a-c \equiv b-d \ (\sim) \\ a \cdot c \equiv b \cdot d \ (\sim) \end{cases}$

$(\forall a, n \in \mathbb{Z},\ a \equiv a \pmod n)$

② $a+(b+c) \equiv (a+b)+c \pmod n$

$a(bc) \equiv (ab)c \pmod n$

$a \equiv b \ (\sim) \Rightarrow b \equiv a \ (\sim)$

$a+b \ (\sim) \equiv b+a \ (\sim)$

$ab \equiv ba \ (\sim)$ → (×)

2

For $\forall a, b, n \in \mathbb{N}$ and $n \geq 2$,

$a \equiv b \pmod{n} \iff r_1 = r_2$

$\begin{cases} a = q_1 n + r_1 \\ b = q_2 n + r_2 \end{cases}$

Thm.: $a \equiv b \pmod{n} \iff n \mid a-b$

$(\iff n \mid b-a)$

$\begin{cases} a \equiv b \pmod{n} \\ c \equiv d \pmod{n} \end{cases} \Rightarrow \begin{cases} a \pm c \equiv b \pm d \pmod{n} \\ ac \equiv bd \pmod{n} \end{cases}$

Proof:

let $\forall k \in \mathbb{N}$

k times $\begin{cases} a \equiv b \\ \vdots \\ a \equiv b \end{cases} \longrightarrow a^k \equiv b^k$

$(1^k \equiv 1$

$(-1)^{2k} = 1, \quad (-1)^{2k+1} = -1$

$(S \equiv k \pmod n) \iff n \mid S-k$ ---①

$22^{49} \cdot 29^6 - 16^2 \pmod 5$ ---②

$16 \equiv 1 \pmod 5$

$16^2 \equiv 1^2 \equiv 1 \pmod 5$

$22 \equiv 2 \pmod 5$

$\Rightarrow 22^{49} = 2^{49}$

$2^2 = 4 \equiv -1 \pmod 5$

$2^4 = (2^2)^2 \equiv (-1)^2 \equiv 1 \pmod 5$

$49 = 4 \cdot 12 + 1$

$2^{49} = (2^4)^{12} \cdot 2 \equiv 2 \pmod 5$

$29 \equiv -1 \Rightarrow 29^6 \equiv (-1)^6 \equiv 1 \pmod 5$

Last digit (mod 10)

$877^{5620146} \equiv 7^{5620146}$

$877 \equiv 7 \pmod{10}$

$7^{4k+r} = (7^4)^k \cdot 7^r$

$\equiv 1^k \cdot 7^r \equiv 7^r$

(反) (反 ~~~)

Thm.: $\forall n, k \in \mathbb{N}$,

$ka \equiv kb \pmod{kn}$

$\Rightarrow kn \mid k(a-b) \Rightarrow n \mid a-b$

$\Longrightarrow a \equiv b \pmod{n}$

E.g.: $42 \equiv 12 \pmod{10}$

$\Leftarrow 3 \cdot 14 \equiv 3 \cdot 4 \pmod{10}$

$\Leftarrow 14 \equiv 4 \pmod{10}$

(but! $7 \not\equiv 2 \pmod{10}$)

(反) 6, 10 ✗ $\gcd(\sim) = 2 > 1$

3, 10 ✓ $\gcd(\sim) = 1$

(一一一必要)

E.g.:

Let $m \equiv 3 \pmod 4$

Prove: $\nexists$ integer solution of $a^2+b^2=m$

Proof:

Assume $\exists a, b, \in \mathbb{Z}$,
s.t. $a^2+b^2=m$

Then $a^2+b^2 \equiv m \equiv 3 \pmod 4$

Let $x \in \mathbb{Z}$,

then

x	0	1	2	3
x^2	0	1	0	1

$\pmod 4$

Let $\forall a, b \in \mathbb{Z}$,

$a^2+b^2 \equiv 0$ or 1 or $2 \pmod 4$

$\not\equiv 3$

↓

contradiction



Greatest Common Divisor

Def.: $\forall a, b \in \mathbb{Z}$ and a, b ~~are~~ not both 0,

$\gcd(a,b) \neq d$),

s.t. $\forall d$, the common divisor of a, b,

$d \geq d_1$,

$\left(\forall d_1, \begin{cases} d_1 \mid a \\ d_1 \mid b \end{cases} \Rightarrow d_1 \mid \gcd(a,b)\right)$

~~$\forall$~~ $\exists$

Bézout identity

$\forall m, n \in \mathbb{Z}$ and not both 0

$\exists x, y \in \mathbb{Z}$, s.t. $\gcd(m,n) = xm + yn$

$(\forall a \neq 0, \gcd(a, 0) = |a|)$

$(a \mid 0 \checkmark \longleftarrow 0 = 0 \cdot a)$

Cor. 3.19: $\begin{cases} \gcd(a,b)=1 \\ a \mid bc \end{cases} \Rightarrow a \mid c$

Proof:

Let $\gcd(a,b)=1$, where $a, b \in \mathbb{Z}$

By Bézout,

$\exists x, y \in \mathbb{Z}$, s.t. $1 = xa + yb$

$a \mid bc \Rightarrow \exists k$, s.t. $bc = ka$

Def.: $n \in \mathbb{N}$, (mod n)

Let $a \in \mathbb{Z}$

$\exists x \in \mathbb{Z}$, s.t. $xa \equiv 1 \pmod{n}$

$\Rightarrow x$ is called a multiplicative inverse

E.g.: 6 has no inverse (mod 10)

$3 \cdot 7 = 21 \equiv 1 \pmod{10}$

3

Bézout identity

$\forall a, b \in \mathbb{Z}$, $\exists x, y \in \mathbb{Z}$,
s.t. $\gcd(a, b) = xa + yb$

Relatively prime (又: coprime)

$\forall$ ~~—~~ ⊙ are r. prime

$\Leftrightarrow \gcd(a, b) = 1$

Lemma:

$$\begin{cases} a \mid bc \\ \gcd(a, b) = 1 \end{cases} \Rightarrow a \mid c$$

~~不~~ 可:

$$\begin{cases} a \mid c \\ b \mid c \\ \gcd(a, b) = 1 \end{cases} \Rightarrow ab \mid c$$

Proof:

(for 又 :)

Assume that $a \mid c$, $b \mid c$ and $\gcd(a, b) = 1$

$a \mid c \Rightarrow \exists k \in \mathbb{Z}$, s.t. $c = ka$

$b \mid c \Rightarrow b \mid ka$

$$\gcd(a,b)=1$$

(lemma) $\Rightarrow b \mid k$

$$\Rightarrow \exists m \in \mathbb{Z},\ s.t.\ k = mb$$

$$\Rightarrow c = ka = mb \cdot a = m \cdot (a \cdot b)$$

$$\Rightarrow ab \mid c$$

$n^2 \not\equiv n \pmod 3 \iff n \equiv 2 \pmod 3$

(R)(L)

Lemma:

$$\gcd(a,n)=1 \Rightarrow \exists\, a^{-1} \pmod n$$

Proof:

$$\gcd(a,n)=1$$

(Bézout) $\Rightarrow \exists x, y \in \mathbb{Z},\ s.t.\ xa + yn = 1$

$\Rightarrow$

$\Rightarrow$

... (2nd)

Let $n=p$ be prime,

$a \equiv 1, 2, \cdots, p-1$ ①

② $(a \not\equiv 0 \pmod p)$

$\forall 1 \le j \le p-1$,

$\gcd(j, p) = 1 \Rightarrow \exists a^{-1} \pmod n$

Assume that $\forall a \not\equiv 0 \pmod n$ has a multiplicative inverse

If $a^{-1} \pmod n$ exists

$\Rightarrow$ then $\gcd(a, n) = 1$

... ①

... ① all are real prime with n

... ①, $\forall j = 1, 2, \cdots, n-1$, ... ②

... ③ $n = k \cdot m$

Let $m < n$

... ④

③ $\gcd(k, n) = k$

$\Rightarrow k = 1$, which contradicts

Hence, n is prime

Assume $\gcd(a,b)=1$

If $a=0 \Rightarrow$ then $b=1$

$\Rightarrow$ then $x \cdot 0 + b \cdot b = 1$

If $a \neq 0 \Rightarrow$ then $b \neq 0$

If $|b|=1$

$\Rightarrow$ then $0 \cdot a + b \cdot b = 1$

If $|b| \geq 2$,

W.L.O.G.,

assume that $b \geq 2$ ($b > 0$)

Assume $-b$ for $b>0$

Let $b<0$,

$\Rightarrow$ then $-b > 0$

$\sim\sim\sim$ (모순)